

Structured summary: | Power Analysis for Longitudinal Clinical Trials with Run-In and Common Close Observations

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This document is a structured, section-by-section summary of the key points argued in the companion manuscript, “[Power Analysis for Longitudinal Clinical Trials with Run-In and Common Close Observations.” It is provided as a supplementary reading aid and does not stand on its own; all notation, derivations, simulation detail, and citations are in the main manuscript.

1 Introduction

Beyond its traditional roles, the run-in period has a distinct methodological value in rate-of-change trials: estimating individual baseline trajectories to raise power.

- Conventional run-in uses include identifying compliant participants and washing out prior treatments.
- When the outcome is a rate of change, run-in observations estimate each participant’s baseline trajectory.
- Incorporating those trajectories into the analysis model substantially increases statistical power.

Frost and colleagues established the mixed-model foundation: run-in observations help most when slope variability is large relative to measurement error.

- The key insight is that pre-randomization observations inform each participant’s underlying trajectory.
- This reduces the residual variance of the treatment-effect estimator, and the gain grows with the slope-to-noise ratio.

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- In Alzheimer’s neuroimaging trials, where that ratio is favorable, they reported sample-size reductions of 30 to 45 percent.

Wang and colleagues extended the framework with two-period models that jointly model run-in and post-randomization data rather than treating run-in as a covariate.

- Joint modelling of both periods yields additional power gains of up to 15 percent.
- It permits unequal randomization ratios without power loss.
- Run-in observations contribute placebo-equivalent information for all participants.

A parallel literature gives closed-form variance and sample-size formulae for the random-coefficient model without a run-in phase, which form the baseline this paper extends.

- Lu and colleagues derived the cLDA estimator variance, its link to ANCOVA, and the conditions under which the two coincide.
- Hu and colleagues generalized these to random-coefficient models with monotone missing data and variable follow-up, illustrated with ADNI data.
- These results cover only the $J_0 = 0$ case and serve as the baseline extended here.

The CRAN longpower package is the reference implementation of the baseline formulae, and our general result reduces to its routines at the appropriate boundaries.

- longpower provides edland, diggle, and liu.liang routines for slope-comparison trials.
- The Stata package slopepower offers a complementary simulation-based route to the same class of calculations.
- Our formula reduces to the Ard-Edland slope variance at $J_0 = J_2 = 0$, and to the Diggle compound-symmetry form when also $\sigma_b^2 = 0$.
- The generalization thus embeds the existing longpower formulae as the baseline against which run-in and common-close gains are measured.

No prior closed-form power formula handles an arbitrary number of run-in observations together with a separate common-close phase, the gap this paper fills.

- The common-close phase observes all participants off-treatment after the randomized period.
- It supplies extra information about individual trajectories at no cost to the treatment comparison.
- Such post-treatment periods appear in delayed-start and randomized-withdrawal designs, but their GLS efficiency contribution has not been formally characterized.

The paper presents three nested results, all derived from the GLS estimator via the Woodbury identity.

- A closed-form variance for exactly two run-in observations (Section 3.2), then its generalization to arbitrary J_0 (Section 3.3).

- A further extension adding J_2 common-close observations (Section 3.4).
- All expressions are in terms of σ^2 , σ_b^2 , and spacing t , with numerical examples from Alzheimer's neuroimaging studies (Section 5).

1.1 General formulation

The model accommodates a two-arm longitudinal trial whose observation schedule spans three distinct phases.

- N participants are observed at equally spaced time points.
- The phases are run-in (pre-randomization), treatment (post-randomization), and an optional common close (post-treatment).
- Y_{ij} denotes the outcome for participant i at time point j .

The outcome model combines a random intercept and random slope with a three-component within-subject variance structure following Frost and colleagues.

- a_i and b_i are the random intercept and slope, while u_{ij} and ε_{ij} are the between-visit and between-scan residuals.
- Both residual terms are independent across visits, so under a single scan per visit they enter every formula only through their sum.
- We therefore define the total within-visit residual $w_{ij} = u_{ij} + \varepsilon_{ij}$ with variance σ^2 .

The two within-visit components are identifiable only with replicate scans, so single-scan designs depend solely on their sum.

- Separating σ_u^2 from σ_ε^2 requires a visit carrying replicate scans.
- For the single-scan designs treated here, only the sum σ^2 enters the treatment-effect variance.
- This reduction simplifies all subsequent variance expressions.

Three fixed-effect slope parameters describe the run-in, placebo post-randomization, and treatment trajectories under equally spaced observations.

- δ is the run-in rate of change common to all participants.
- β is the placebo post-randomization rate of change, and γ is the treatment effect on that rate.
- Observations are equally spaced with interval t , so $t_j = jt$.

During the common close both arms revert to a shared off-treatment slope, modelled by deactivating the treatment indicator in that phase.

- For $j > J_1$ all participants are off-treatment with slope $\beta + b_i$.
- This is implemented by setting $g_i = 0$ for every participant during the common close.
- Equivalently, the treatment indicator is replaced with $g_i \cdot I(1 \leq j \leq J_1)$ in the model.